

Small Is Beautiful: Geometric Composition for CPU-Only Document Conversion at 251 MB

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Abstract

Document parsing is dominated by billion-parameter vision-language models and multi-gigabyte GPU pipelines. We present PRISM, a parser whose entire model inventory is 251 MB and which runs exclusively on CPU at 5.9 s per page median. We compose four small specialists (layout with native reading order, lightweight OCR, a 20M-parameter formula recognizer, a 7 MB table model) with deterministic geometric reasoning: formula band-splitting, fraction-bar fusion and spanning-cell assembly. Where a component has a known blind spot, the alternative path is taken only after a cheap recognition probe confirms that the page reads better. The geometry supplies accuracy (Character Detection Matching on display formulas rises 56.9 → 78.1 with zero added parameters) and the probes supply robustness out of distribution. Applied to camera-capture normalization it yields a finding of independent interest: modern OCR is nearly invariant to photometric defects, so correction should be a verification-gated damage filter rather than a default enhancement stage. On OmniDocBench v1.6 (1651 pages, official harness) PRISM scores 87.3. We reran the MinerU 3.4.4 pipeline backend CPU-only ourselves on the same machine and harness commit, as no independently published CPU-only MinerU number exists: PRISM leads this self-measured baseline by 1.2 Overall and 6.3 CDM at 4.8× fewer weight bytes, with a 6× smaller p99 latency tail and 2.5 GB less peak RAM. PRISM also scores 89.0 on the v1.5 cut, above every pipeline system reported there. We report ablations for every design decision and a perturbation audit of the benchmark itself; a stage-by-stage decomposition attributes two thirds of PRISM’s gain over its own baseline to deterministic orchestration rather than model capacity.

1. Introduction

We address image-to- $\text{\LaTeX}$ document conversion: given a scan or a photograph of a page, recover its body text in read-

ing order, its formulas as compiling $\text{\LaTeX}$ and its tables as HTML. The $\text{\LaTeX}$ target is what makes the task hard under a small-model budget, and it is where most of our gain lands. Retrieval-augmented generation, digital archiving, accessibility tooling and foundation-model corpus construction all depend on it. The current state of the art is held by specialized vision-language models (VLMs) of 0.9B–241B parameters [6, 12, 32], and by GPU pipeline systems such as MinerU [44] and Marker [36] that chain multi-hundred-megabyte detectors and recognizers, typically including a VLM component for OCR or formulas. Both families presuppose datacenter or at least discrete-GPU deployment.

$\text{\LaTeX}$ emission is required because mathematical structure is not representable in plain text; a fraction, a matrix or a limit collapses into unordered characters without it. Formula conversion is also the component under the greatest pressure from a small-model budget, since a 20M-parameter decoder must produce an exactly-compiling symbolic string rather than an approximately-correct one. Most of our design effort, and most of our measured gain, goes there. Many deployments cannot meet that presupposition: on-premise processing of confidential documents, batch digitization on office hardware, offline edge deployment, and cost-sensitive high-volume ingestion. For these the question is not the best score achievable but how much of it survives a hard budget of no GPU, a few gigabytes of RAM, seconds per page.

PRISM answers this under three simultaneous constraints: CPU-only execution, model weights under 300 MB, and single-digit seconds per page. It contains no generative model larger than 20M parameters; its accuracy comes from an orchestration layer around four small specialists: a layout detector that also predicts reading order (RT-DETR, 130 MB), a text OCR stack (31 MB), a distilled formula recognizer (Texo, 20M), and a table model (SLANet-plus, 7.4 MB). Deterministic geometry does what end-to-end systems leave to attention: formula band-splitting, fraction-bar fusion, dense-grid detection, spanning-cell resolution, and recognition-verified nor-

malization (Sec. 3.1). Input is not restricted to clean scans: photographs of pages enter through the same verified path.

On OmniDocBench v1.6 [35] (1651 pages, ten document types, EN+ZH, official harness) PRISM reaches **87.3** in 1.7GB peak RAM: above the 7B GPU VLM class (olmOCR 85.7, Mistral OCR 85.7) and past a MinerU pipeline backend we re-ran ourselves under identical CPU conditions (+1.2 Overall, +6.3 CDM), a margin an order of magnitude above the convention sensitivity we measure in Sec. 4.8. On the v1.5 cut it scores **89.0**, above MonkeyOCR-pro-3B, dots.ocr and Qwen2.5-VL-72B.

1.1. Novelty

We substitute deterministic ink-mass geometry for recognizer capacity and replace input-statistics thresholds with probe-verified routing (*algorithmic*). This is delivered as a six-graph, 251MB, fp32, CPU-only inventory with no framework runtime resident (*system*), and is targeted at offline and on-premise conversion of scans and page photographs to $\text{\LaTeX}$ where GPU deployment is unavailable (*application*).

Our contributions are:

- **A complete, reproducible recipe** for CPU-only image-to- $\text{\LaTeX}$ conversion at 251MB reaching 7B-VLM-class accuracy, validated end-to-end on the official OmniDocBench harness.
- **Deterministic image-to- $\text{\LaTeX}$ recovery** that substitutes for model capacity and supplies the system’s largest gains. Ink-mass grouping recovers formula structure geometrically (display-formula CDM 56.9 $\rightarrow$ 78.1 with zero added parameters, +8.09 Overall), alongside detector-native reading order, where better detection is found to make geometric ordering *worse*, and spanning-cell table assembly.
- **Recognition-verified normalization as a finding:** corrections applied as *proposals* gated by the pipeline’s own 9.9MB text-detection graph, at zero weight cost. The study inverts the conventional design: modern OCR is nearly invariant to photometric defects, and the correct policy is a gated damage filter whose dominant action is rejection.
- **A perturbation audit of the benchmark itself:** re-scoring semantically-null transformations of our predictions shows the harness robust to styling but moved 1.54 reading-order points and 0.14 composite points by segmentation granularity alone (Sec. 4.8). Sub-point component gaps reflect convention as much as capability.
- **An efficiency frontier study** placing pipelines and small VLMs on identical hardware across accuracy, latency, peak RAM and model size.

2. Related Work

Document parsing pipelines. Classical document AI decomposes the task into layout analysis [16, 37, 51], OCR [40, 41], specialist recognition [11, 43], and reading-order recovery [15]. Modern exemplars are CPU-*capable* but engineered for GPUs: MinerU [44, 46], Marker [36] (a $\sim$ 650M Surya OCR VLM), Docling [1] and PP-Structure V3 [7]. On CPU we measure PP-StructureV3 at 176s/page and 5.7GB in its server configuration (Tab. 3). PRISM is the first pipeline designed *from the start* for a strict CPU/memory/size budget and evaluated end-to-end on the full official benchmark.

Vision-language models for documents. The OCR-free line begun by Donut [18], Pix2Struct [19] and Kosmos-2.5 [26] has matured into specialized document VLMs that now top every benchmark: PaddleOCR-VL (0.9B) [6], MinerU2.5 (1.2B) [32], GLM-OCR [12], dots.ocr [22], MonkeyOCR [23], DeepSeek-OCR [49], Vary [47], and general VLMs such as Gemini [5] and Qwen-VL [2, 3]. Their accuracy is bought with GPU inference. The “efficient” end of this family, SmolDocling-256M [31] and GraniteDocling-258M [17], matches PRISM’s *size* class but not its efficiency class: as autoregressive generators they emit the entire page token by token, costing 84–130s per page on CPU (Tab. 3), and their accuracy trails classical pipelines. Nougat [4] and GOT-OCR2.0 [48] are earlier single-model parsers with similar deployment profiles.

Component models. PRISM composes open components: PP-DocLayoutV3 [8, 42] (RT-DETR [50] with native reading order), PP-OCR [7, 21] detection [24] and recognition, the Texo distillation [28], and SLANet-plus [20]. So do MinerU, Marker, Docling and PP-Structure V3, none of which report how their accuracy divides between the components and the composition around them. Figure 4 supplies that decomposition on the full official benchmark.

Document image enhancement. Prior correction of captured documents is uniformly *open-loop*, consuming the correction unconditionally: adaptive binarization [39], learned unwarping [10, 27] and joint geometric-photometric restoration [13]. Section 4.5 shows why this underdelivers; PRISM’s probe-verified proposals are, to our knowledge, the first closed-loop formulation evaluated on an end-to-end parsing benchmark.

Benchmarks. OmniDocBench [35] is the standard end-to-end benchmark: ten genres, bilingual, component-wise metrics (normalized edit distance for text and order, Character Detection Matching (CDM) [45] for formulas, Tree-Edit-Distance Similarity (TEDS) [52] for tables), official

matcher-based harness. We target v1.6 (1651 pages), report the v1.5 cut for comparability, and use olmOCR-Bench [38] and Fox [25] as generalization checks under orthogonal scoring.

3. Method

Problem formulation. Given a page image I , we recover an ordered document $D = (y_{\pi(1)}, \dots, y_{\pi(n)})$. A detector f_θ supplies regions and their reading order, $f_\theta(I) = \{(b_i, c_i)\}_{i=1}^n$ with permutation π , where b_i is a box and $c_i \in \mathcal{C} = \{\text{TEXT}, \text{FORMULA}, \text{TABLE}, \dots\}$. Each class carries an emission grammar $\mathcal{L}_c$: L^AT_EX for FORMULA, HTML for TABLE, plain text otherwise. A specialist g_c maps a crop to $y_i \in \mathcal{L}_{c_i}$.

The naive composition decodes each detected box directly, $y_i = g_{c_i}(I|_{b_i})$, which fails whenever the detector’s geometry and the recognizer’s expected input disagree. PRISM inserts a deterministic map T_c that rewrites one crop into a sequence of recognizer-appropriate crops, $T_c(x) = (x^{(1)}, \dots, x^{(m)})$, and reassembles the decodings under a grammar-aware operator $\oplus_c$:

$$y_i = \bigoplus_{c_i} g_{c_i}(T_{c_i}(I|_{b_i})). \quad (1)$$

T_c is defined by ratios of measured ink geometry (Sec. 3.3) and introduces no parameters.

Verification gating is the second operator. Let Φ be a family of candidate input transforms and p a cheap probe scoring the confident-text surface of a crop. Rather than threshold on input statistics, we accept a transform only on measured evidence:

$$\hat{x} = \begin{cases} \varphi(x), & p(\varphi(x)) \geq \tau p(x), \\ x, & \text{otherwise,} \end{cases} \quad \varphi \in \Phi, \quad (2)$$

with $\tau = 1.02$ throughout. Because p reuses a graph already in the inventory, Eq. (2) costs no additional weights.

The system optimizes D under three simultaneous constraints:

$$\sum_\theta |\theta| \leq B, \quad \text{device} = \text{CPU}, \quad \mathbb{E}[t_{\text{page}}] < 10 \text{ s}, \quad (3)$$

with $B = 300 \text{ MB}$. We claim that under Eq. (3), most of the achievable accuracy comes from T_c and $\oplus_c$ in Eq. (1) rather than from the capacity of g_c , and that Eq. (2) is what makes the remaining components safe out of distribution (Sec. 4.6).

We compose four small specialists into a five-stage pipeline (Fig. 1). A page is processed by: (1) layout detection with native reading order, (2) recognition-verified per-region preprocessing, (3) specialist recognition of text, formulas and tables, (4) deterministic reading-order assembly, and (5) emission of benchmark-compatible Markdown and HTML. All models run as ONNX graphs on CPU in persistent worker subprocesses, so no PyTorch or Paddle runtime is resident and models load once.

3.1. Input normalization

Scanned or born-digital pages pass through untouched except for a resolution cap. An entropy and occupancy modality classifier, refined by pure-white fraction, identifies camera captures. These receive a correction stack: specular-glare inpainting, illumination flattening by background division, perspective rectification recovering up to 25° of page rotation and off-axis capture, and conditional contrast enhancement.

Recognition-verified corrections. We must decide *when* each correction applies, which proves harder than designing it. Every pixel-statistics discriminator we measured gave inseparable distributions on real shadowed captures versus clean pages: slide gradients score *higher* on shadow statistics (0.60) than genuine shadows (0.06–0.29), so any open-loop threshold either misses defects or corrupts clean pages. PRISM instead treats each correction as a *proposal*: the pipeline’s own 9.9 MB PP-OCRv6 detection graph (a DB-Net [24]), re-run once per proposal at 640 px ($\sim 120 \text{ ms}$) on camera captures only, scores the confident-text surface before and after, and the proposal is kept only if the score clearly improves ($\geq 2\%$). The verifier rejects far more often than it accepts, and our measurements show this is the correct default (Sec. 4.5). The gate also neutralizes two failure modes intrinsic to the correction stack: a shadow divide whose background estimate contains the text ink, rejected at $7\times$, and glare inpainting that smears bright paper.

3.2. Layout detection

A single PP-DocLayoutV3 detector [8, 42] (the RT-DocLayout release of the PP-DocLayout line; RT-DETR with a mask-based head, 800 px, 130 MB ONNX) provides all regions *and* a native reading-order index per box, which we exploit directly (Sec. 3.5). Against its box-only predecessor (PP-DocLayout-plus-L, identical size class) this detector is the largest single model contribution in Fig. 4, driven almost entirely by dense stock-listing pages where the predecessor detects 13 of ~ 80 agate tables and V3 detects 78. Detector ensembling is net negative under this budget: a YOLO math-formula detector raised formula-page CDM by 3.3, but its false positives cost more text-edit than that gained. We instead invest in class-aware post-processing:

• **Per-class confidence gates.** Formulas are accepted at confidence ≥ 0.30 , other classes at ≥ 0.50 : formula false negatives are catastrophic (the region is OCR’d as garbled text) while false positives are cheap. Sweeping the formula gate across $[0.20, 0.40]$ moves Overall by 0.01: a broad plateau, not a tuned peak. Lowering the table gate to 0.30 admits detections costing more text-edit than the recovered tables gain.

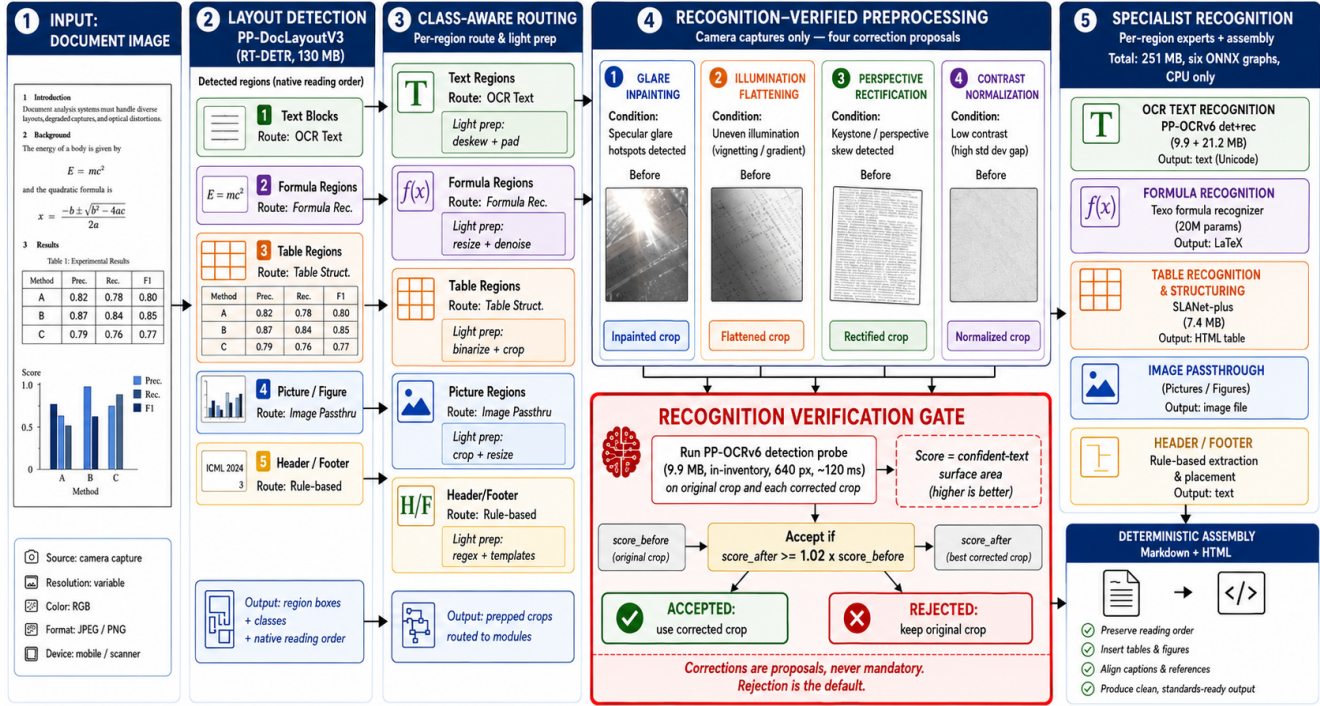


Figure 1. PRISM pipeline and the recognition-verification gate. A single layout detector identifies document regions and their native reading order. Each region passes through recognition-verified preprocessing, where photometric corrections are proposals accepted only if a lightweight probe improves confidence. Verified crops go to specialist recognizers for text, formulas and tables before deterministic assembly produces Markdown and HTML.

- **Containment resolution with exemptions.** Overlapping boxes are resolved by containment suppression, with three class-conditional exemptions: a Formula inside a Text block is never consumed (the text block is masked instead); a near-page-sized Picture (>70%) never consumes others (usually background); and an outer Formula containing ≥ 2 higher-confidence Formulas loses to the chips.

3.3. Formula processing by ink-mass geometry

The formula recognizer is Texo [28], a 20.4M-parameter distilled encoder-decoder (81.9 MB ONNX, encoder plus decoder). We find that cropping each detected Formula box and decoding it directly is the single largest error source in the naive pipeline, and that the cause is a mismatch between the detector’s box geometry and the recognizer’s expected input rather than recognition itself. A deterministic geometric layer (“formula v2”) is therefore inserted between detection and recognition. Its predicates are ratios of measured ink geometry rather than absolute thresholds, so they are invariant to resolution, font size and crop scale.

- **Row-band splitting.** A detected box often covers a multi-line derivation. We binarize ink (Otsu [34]) and project onto rows: a gap exceeding a quarter of the band height is inter-line leading, not intra-expression spacing,

a separation that holds across typesetting systems. The rule is suspended where a vertical ink bridge spans the gap, since continuous ink across a line break is the defining signature of a matrix or a tall delimiter, which must stay whole.

- **Fusion and trimming.** A thin band wider than both neighbors is a fraction bar, and numerator, bar and denominator are re-fused into one expression. Horizontal gaps wider than the band is tall separate expressions rather than terms, and a short trailing chunk set off from the body is an equation number and is trimmed.
- **Inline-guard with dense-grid exemption.** A Formula box whose row also contains substantial non-formula ink inside an enclosing Text region is inline math and is left to OCR, since decoding it as display math double-emits it. Ink from *other* formula boxes is blanked first, and a Text region with ≥ 4 formulas holding $\geq 50\%$ of its ink is a math dense grid and is exempt from the guard; both conditions are required.
- **Inline-math splicing.** A guarded inline chip is recognized by the formula model and spliced into the host block as an inline $\$. . . \$$ span at the character position interpolated from its horizontal centroid; the host is re-OCR’d with chips masked. Spans decoding to structural

output or CJK fall back to masked OCR.

- **Detector class recovery.** PP-DocLayoutV3 labels handwritten and standalone formulas `inline_formula`; a pipeline that drops that class silently loses every formula on handwritten-notes pages (near-zero-CDM formulas: 20 with routing vs. 77 without). Routing the class into the Formula path, with the inline guard discarding genuine in-text chips, moves display CDM $85.4 \rightarrow 87.6$ with academic control pages unharmed.
- **Out-of-distribution escape hatch.** Texo is trained on English math and substitutes unrelated symbols when a formula embeds CJK text (0.65 mean CDM on Chinese-page formulas vs. 0.87 English). On CJK pages each formula crop is probed with the line OCR; if it reads two or more confident CJK characters, PRISM emits OCR-derived $\text{\LaTeX}$ (CJK runs wrapped in `\text{}`, rows joined as an array) instead of decoding at all. This is recognition-verified routing, as in Sec. 3.1.

This layer moves display-formula CDM from 56.9 to 78.1 with zero new parameters and no measurable latency cost, the largest single gain in the system. The gain holds across every OmniDocBench document type and language stratum (Suppl. Tabs. S1–S2) and transfers to two benchmarks with different provenance and scoring (Sec. 4.4). Suppl. Fig. S2 shows the four stages operating on a single detected box.

3.4. Tables

Table structure and cell content are predicted in a single pass by SLANet-plus (7.4MB), a later PaddleX release of the SLANet architecture [20] over the detected table crop. It replaces a detection-based table transformer with externally assigned OCR tokens, whose row proposals shattered on dense tables: TEDS $71.4 \rightarrow 78.8$, catastrophic tables (TEDS < 0.3) recovering from 0.01 to 0.57.

3.5. Reading order and assembly

We emit blocks in the detector’s native reading order. An order index per box is predicted by PP-DocLayoutV3 in the same forward pass, and synthetic regions (rescued text, split boxes) inherit their nearest ordered neighbour’s index. Two measurements support this choice. The detector’s order halves reading-order edit distance against our strongest geometric stack (column-aware sort, DAG splitting, recursive XY-cut [15]): $0.353 \rightarrow 0.171$ on the development subset, 0.147 full-benchmark. Less obviously, the geometric stack *inverts* the value of better detection, degrading to 0.399 with V3’s finer boxes because more correct boxes give a gap heuristic more chances to interleave columns, so it is retained only as a fallback. Assembled regions are lowered to OmniDocBench-style Markdown with math and HTML tables quarantined from text-mode rewrites; without that quarantine, 28% of matched formula pairs are silently

corrupted.

3.6. Global guards

Two cheap guards bound worst-case output. If the assembled body contains fewer than 30 content characters, the full normalized page is OCR’d as a single block (rescuing stylized covers and pages where layout detection collapses); and near-page-size Formula/Picture detections are dropped before containment resolution, since undropped they consume every real region on the page and emit an empty document.

4. Experiments

4.1. Setup

Benchmark. OmniDocBench v1.6 [35]: 1651 pages across ten genres in English and Chinese. We run the *official* harness unmodified (quick-match matcher; text and order by normalized edit distance, display formulas by Character Detection Matching (CDM) [45], tables by Tree-Edit-Distance Similarity (TEDS) [52]), with the leaderboard composite Overall = $[(1 - \text{TextEdit}) \cdot 100 + \text{CDM} + \text{TEDS}] / 3$, and additionally report the v1.5 cut. Design decisions were made on internal validation subsets; all full-benchmark numbers here and in the ablations are confirmatory re-runs, and the zero-shot olmOCR-Bench and Fox evaluations involve no configuration selected against them.

Hardware. All PRISM measurements: an Intel Core i7 (8 physical / 16 logical cores) consumer Windows 11 machine, 32 GB RAM, CPU only, onnxruntime 1.x; no additional quantization applied. Latency is end-to-end per page including all workers; peak RAM is the maximum working set of the whole process tree.

4.2. Main results

Table 1 places PRISM on the v1.6 leaderboard. It lands above the 7B VLM class (olmOCR, Mistral OCR), exceeds Marker by 8.9 points, and overtakes the MinerU pipeline backend (87.29 vs. 86.08) on the composite and on formulas (+6.3 CDM). A paired bootstrap over pages (10,000 resamples, harness aggregation replicated exactly) gives +1.21 Overall (95% CI [0.45, 2.35]) and +6.33 CDM (CI [3.34, 9.45]). The re-run reproduces MinerU’s published figure to within 0.4 Overall. PRISM leads it at $4.8\times$ fewer weight bytes (251 MB vs. 1.2 GB), with a p99 latency tail $6\times$ smaller (20.0 vs. 124.3 s/page) and 2.5 GB less peak RAM (1.70 vs. 4.25 GB): the larger pipeline degrades on CPU precisely on formula- and table-heavy pages, the ones document parsing exists for; it retains an edge on plain text, purchased with a 776 MB recognizer stack.

Figure 2 shows representative PRISM outputs across formula, table, multi-column and mixed-script pages.

formula

CDM 99.6

raw page (crop)

For consistency, the time derivative of the constraints of (10) must vanish and hence they must have vanishing Poisson bracket with H . Using the fundamental Poisson brackets

$$[U(x), \Pi^U(y)] = \delta(x - y),$$

etc., we find that the primary constraints of (10) imply the secondary constraints

$$(\Sigma, \Sigma_i) = (-\partial_k \Pi_k^U, e^{ij} \partial_j (\Pi_k^U - m V_k) - \mu^2 B_i).$$

table

TEDS 94.7

multi-col

order 0.000

mixed

text 0.000

PRISM output, compiled

For consistency, the time derivative of the constraints of (10) must vanish and hence they must have vanishing Poisson bracket with H . Using the fundamental Poisson brackets

$$[U(x), \Pi^U(y)] = \delta(x - y),$$

etc., we find that the primary constraints of (10) imply the secondary constraints

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表 7：11 月小厨电天猫京东各品牌零售端表现

	品牌	销售额同比			销售额占市占率同比 (pct)		
		2023M11	2023 累计	2023M11	2023 累计	2023M11	2023 累计
线上	美的	-2.5%	-0.7%	13.9%	11.6%	2.50	4.25
	九阳	-27.8%	-24.0%	6.4%	6.3%	-0.68	1.25
	苏泊尔	-20.9%	-17.3%	8.8%	9.5%	-0.08	2.44
	小熊	-35.9%	-6.4%	3.1%	3.4%	-0.75	1.00

EYDO. The BLM will not accept personal checks or other non-certified funds.

col 2 (end)

Arrangements for electronic fund transfer to the BLM for payment of the balance due must be made a minimum of two weeks prior to the payment date.

col 3 (start)

参选价值 Why participate

所有优秀参选企业无论是否获奖都将有所收获; BCM program is highly beneficial to all participants!

PRISM output, compiled

For consistency, the time derivative of the constraints of (10) must vanish and hence they must have vanishing Poisson bracket with H . Using the fundamental Poisson brackets

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表 7：11 月小厨电天猫京东各品牌零售端表现

品牌	品牌	销售额同比	销售额环比	销售额环比	销售额环比	销售额环比
美的	美的	-2.5%	-0.7%	13.9%	11.6%	2.50
九阳	九阳	-27.8%	-24.0%	6.4%	6.3%	-0.68
苏泊尔	苏泊尔	-20.9%	-17.3%	8.8%	9.5%	-0.08
小熊	小熊	-35.9%	-6.4%	3.1%	3.4%	-0.75

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Figure 2. End-to-end PRISM output on four OmniDocBench v1.6 pages spanning distinct structures: a display-formula derivation (CDM 99.6), a dense bilingual retail table (TEDS 94.7), a multi-column page linearized across the column break (order 0.000), and a mixed Chinese–English heading block (text 0.000). Left: raw crop. Right: compiled output. All are the submitted build’s actual predictions.

Table 1. OmniDocBench v1.6.full (1651 pages), official harness. Leaderboard rows from the official repository; PRISM measured on the identical dataset and harness. Weights are inference bytes (fp16 for open vision–language models, stack sizes for pipelines); (G) and (C) mark the device on which the row was measured. Every system above PRISM in this table requires a discrete GPU or a paid API. Full per-section breakdown in Suppl. Tabs. S1–S3.

Model	Weights	Overall↑	Text↓	CDM↑	TEDS↑	Order↓
olmOCR [38]	14 GB	85.74	0.139	88.10	83.00	0.216
dots.ocr [22]	6 GB	90.77	0.048	89.95	87.18	0.138
Marker 1.8.2 [36]	~3 GB (G)	78.44	0.157	85.24	65.77	0.243
MinerU2.5-Pro [46]	2.4 GB	95.69	0.036	97.45	93.42	0.120
MinerU-Pipe. [§] [44]	~2 GB (G)	86.47	0.055	83.07	81.88	0.153
PaddleOCR-VL [6]	1.8 GB	94.18	0.040	95.91	90.65	0.135
MinerU2-pipe. [‡] [44]	1.2 GB (C)	86.08	0.056	81.82	82.02	0.153
Gemini 3 Pro [14]	API	92.91	0.064	95.99	89.15	0.165
GPT-5.2 [33]	API	86.59	0.114	88.21	82.95	0.193
Mistral OCR [29]	API	85.66	0.097	89.91	76.78	0.171
PRISM (ours)	0.25 GB (C)	87.29	0.084	88.14	82.15	0.147

[§]as published by MinerU. [‡]mineru 3.4.4 (latest release), pipeline backend in the shipped default configuration (MinerU-LayoutV2, PP-OCRv6, UnimerNet-small, SLANet-plus; component selection is MinerU’s, unmodified by us), re-run CPU-only on the same machine and harness commit as PRISM.

Table 2. OmniDocBench v1.5 cut. Metrics, as in Tab. 1: Text and Order are normalized edit distances against the reference (lower better); CDM is Character Detection Matching on display formulas and TEDS is Tree-Edit-Distance Similarity on tables (higher better); Overall = $[(1 - \text{Text}) \cdot 100 + \text{CDM} + \text{TEDS}] / 3$. PRISM is scored against the official v1.5 annotations and harness, not by re-scoring the v1.6 run; reference rows are as published in the GLM-OCR report. Weights are fp16 bytes for open models.

Model	Weights	Overall↑	Text↓	CDM↑	TEDS↑	Order↓
Qwen2.5-VL-72B [3]	144 GB	87.02	0.094	88.27	82.15	0.102
MonkeyOCR-pro-3B [23]	7.4 GB	88.85	0.075	87.25	86.78	0.128
dots.ocr [22]	6 GB	88.41	0.048	83.22	86.78	0.053
Deepseek-OCR [49]	6 GB	87.01	0.073	83.37	84.97	0.086
Nanonets-OCR-s [30]	6 GB	85.59	0.093	85.90	80.14	0.108
Marker-1.8.2 [36]	~3 GB (GPU)	71.30	0.206	76.66	57.88	0.250
MinerU2.5 [32]	2.4 GB	90.67	0.047	88.46	88.22	0.044
GLM-OCR [12]	1.8 GB	94.62	0.040	93.90	93.96	0.044
PaddleOCR-VL-1.5 [9]	1.8 GB	94.50	0.035	94.21	92.76	0.042
PP-StructureV3 [7]	~1 GB (GPU)	86.73	0.073	85.79	81.68	0.073
Gemini-3 Pro [14]	API	90.33	0.065	89.18	88.28	0.071
Gemini-2.5 Pro [5]	API	88.03	0.075	85.82	85.71	0.097
GPT-5.2 [33]	API	85.50	0.123	86.11	82.66	0.099
PRISM (ours)	0.25 GB (CPU)	88.97	0.077	89.56	85.02	0.138

4.3. Efficiency frontier

On the full-benchmark run (Suppl. Tab. S7) PRISM’s median latency is 5.9 s/page (mean 7.2; p90 12.7, p99 20.0) with a 1.70 GB peak working set. A speed-optimized configuration (83.55 Overall) runs at 3.0 s median, so the last +3.7 points cost +2.9 s. The p99 tail is 3.4× the median, dominated by dense newspaper spreads and formula-heavy academic pages. Table 3 is the same-machine comparison on one fixed page set. PRISM is the fastest and smallest system there and leads on table structure by at least 7.2 TEDS.

Stacks that beat it on plain text and isolated formulas do so at 2–21× the latency and up to 5.5× the memory. This subset scores formulas by raw edit distance on 20 English pages, which penalizes emission convention directly; under CDM on the full benchmark, which normalizes rendering, PRISM leads the same MinerU build by 6.3 points (Tab. 1). The full inventory is six ONNX graphs totalling 251 MB, none exceeding the 130 MB detector (Suppl. Tab. S6), with no PyTorch or Paddle runtime resident.

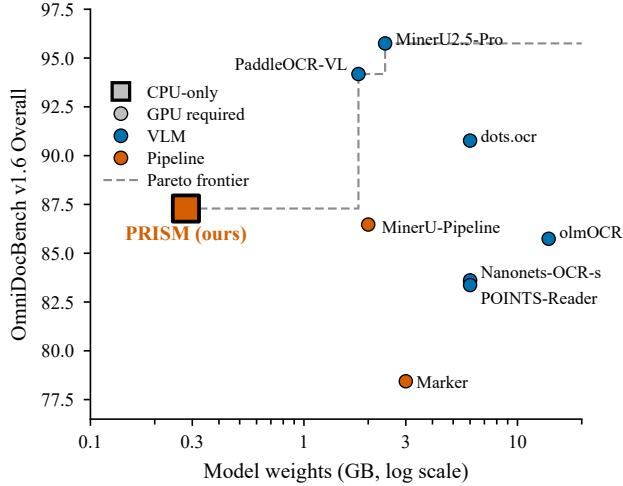


Figure 3. Efficiency frontier on OmniDocBench v1.6. PRISM is the only CPU-only system in the comparison; every system plotted at or above its accuracy is evaluated on a discrete GPU. Plotted systems are MinerU2.5-Pro [46], PaddleOCR-VL [6], dots.ocr [22], MinerU [44], olmOCR [38] and Marker [36]. API-only models are excluded as they publish no weight figure.

Table 3. Controlled CPU head-to-head on one fixed page set: 100 English v1.6 pages selected deterministically for table and formula coverage, identical machine, single-worker 8-thread budget (distinct from the dual-worker, 16-logical-core configuration used for the full-benchmark headline in Tab. S7), mean of two runs. RAM is peak process-tree resident set in GB. Latencies here exceed the 5.9 s/page headline and are comparable across rows only.

System	RAM↓	s/pg↓	Text _{EN} ↓	Formula _{EN} ↓	TEDS↑
PP-Struct.V3 (server) [7]	5.74	176.4	0.055	0.130	71.0
PP-Struct.V3 (mobile) [7]	4.60	18.7	0.141	0.331	51.5
MinerU2-pipe. (3.4.4) [44]	3.12	18.0	0.031	0.178	74.2
GraniteDocling-258M [17]	1.92	83.6	0.172	0.183	68.4
SmolDocling-256M [31]	1.81	129.7	0.463	0.617	45.5
PRISM (ours)	1.04	8.6	0.084	0.218	81.4

Is PRISM GPU-bound? No. A GPU cuts the layout pass 826 → 86 ms, but end-to-end latency is governed by the formula decoder, which is faster per token on CPU. Latency scales 23.1→7.6 s/page from 1 to 12 cores, most of it by 4 cores (11.6 s/page), so it runs on commodity hardware (Suppl. Fig. S4).

Deployment envelope. The target is a workstation that already exists in the building. On a 4-core office machine at 11.6 s/page, a single unattended overnight run clears roughly 2,500 pages, and a 1.7 GB working set leaves an 8 GB desktop usable for other work. That covers the volumes typical of an academic library digitizing a journal backfile, a hospital records office converting scanned charts, or a municipal archive processing planning documents, all settings where the pages cannot leave the premises and no

Table 4. olmOCR-Bench [38] pass rate (% , higher is better) by category. AR = arXiv math, OSM = old-scans math, Tab = tables, OS = old scans, HF = header/footer removal, MC = multi-column reading order, LTT = long tiny text, Base = per-document baseline quality checks. Overall is the pass rate over all assertions with a 95% bootstrap CI. PRISM is zero-shot on the identical build.

Model	AR	OSM	Tab	OS	HF	MC	LTT	Base	Overall↑
olmOCR [38]	74.9	71.2	71.0	42.2	94.5	78.3	73.3	98.3	75.5±1.0
Mistral OCR	77.2	67.5	60.6	29.3	93.6	71.3	77.1	99.4	72.0±1.1
Marker [36]	76.0	57.9	57.6	27.8	84.9	72.9	84.6	99.1	70.1±1.1
GPT-4o	53.5	74.5	70.0	40.7	93.8	69.3	60.6	96.8	69.9±1.1
Qwen2.5-VL [3]	63.1	65.7	67.3	38.6	73.6	68.3	49.1	98.3	65.5±1.2
Gemini Flash 2 [5]	54.5	56.1	72.1	34.2	64.7	61.5	71.5	95.6	63.8±1.2
MinerU [44]	75.4	47.4	60.9	17.3	96.6	59.0	39.1	96.6	61.5±1.1
GOT-OCR2.0 [48]	52.7	52.0	0.2	22.1	93.6	42.0	29.9	94.0	48.3±1.1
PRISM (ours)	56.0	34.9	77.3	20.5	77.0	64.4	69.0	96.2	62.1±1.1

GPU budget exists. The alternative under those constraints is not a larger model but a paid API the documents are not permitted to reach.

4.4. Generalization: olmOCR-Bench

OmniDocBench and Fox both score against a reference transcription, so we validate under scoring that no output convention can satisfy. olmOCR-Bench [38] replaces edit distance with 7,010 binary unit tests on string presence, header removal, block order, cell adjacency and KaTeX render-equality. We run PRISM zero-shot, 200 DPI, official harness.

Table 4 reports per-category pass rates on the current public release (commit 54a96a6; 7 categories, 7,010 tests plus 1,403 baseline checks). The systems above it are billion-parameter VLMs at 24–56× the weight budget. The two benchmarks measure different quantities: OmniDocBench grades average reconstruction continuously, where a character slip is near-free; olmOCR-Bench grades the tail with binary assertions, where one wrong character fails a test outright. The per-category profile follows the design. Transfer is strongest where the orchestration layer determines the output, on tables (77.3, above MinerU’s 60.9), multi-column order (64.4 vs. 59.0) and long tiny text (69.0 vs. 39.1), and weakest where recognition capacity binds, on old scans (20.5) and math (arXiv 56.0, old-scans-math 34.9), where a symbol slip failing KaTeX render-equality is unrecoverable. A direct audit finds 96% of the multi-column and tiny-text failures are genuine recognition misses, not ordering errors; Sec. 5 localizes these.

Fox. On Fox [25] (212 pages, EN+CN) the identical build reaches 0.075 (EN) / 0.093 (CN) edit distance on CPU, between Qwen-VL-Max and Vary-toy, and is scored only after stripping structural markup the baselines never produce (Suppl. Tab. S4).

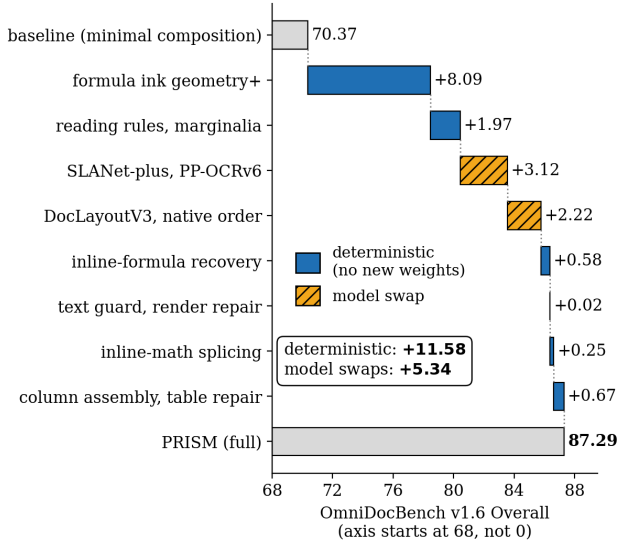


Figure 4. Decomposition of PRISM’s gain over a minimal-composition baseline, official v1.6 harness, identical dataset and scoring throughout. Components that add no new weights (solid) contribute +11.58 Overall; model choices (hatched) contribute +5.34. Deterministic orchestration, not model capacity, supplies more than two thirds of the gain.

4.5. Recognition-verified normalization

We evaluate the gate on captures and degradations, where it operates, rather than on the benchmark, which is dominated by pages that bypass it (Suppl. Tab. S8). We find that **modern OCR is largely invariant to photometric page defects**. A $0.18\times$ shadow raises PP-OCrv6’s block edit distance by only 0.004; in four of five degradations the uncorrected image is at or near best, and open-loop correction taxes the clean case ($0.177 \rightarrow 0.184$). The gate’s rejection rate falls monotonically with severity (75.9% clean to 45.2% under combined shadow and cast), opening precisely when defects are real, and no fixed pixel-statistics threshold reaches this operating point (Sec. 3.1).

Photometric normalization belongs in the pipeline as a *verification-gated damage filter* rather than a default enhancement stage. The stack itself stays, since the same pass rectifies off-axis capture. On 44 uncontrolled captures the open-loop stack is significantly worse than no correction (-0.095 , 95% CI $[-0.166, -0.040]$), while the gate recovers that loss ($+0.091$, CI $[0.037, 0.160]$) and is indistinguishable from doing nothing (-0.004 , CI $[-0.011, +0.002]$), bounding its cost at one point (Suppl. Tab. S9).

4.6. Ablations

Figure 4 decomposes PRISM’s gain over a minimal pipeline baseline (the same four specialist stages with naive com-

position: whole-box formula crops, geometric reading order, no orchestration layer) into its constituent design decisions. Each bar is a leave-one-out delta: the component is removed from the otherwise-full system and the full v1.6 benchmark is re-scored, so bars are order-independent and are rendered cumulatively for readability only. Components adding no new weights contribute +11.58 Overall, model choices +5.34. The largest single contribution is deterministic, the formula-geometry layer of Sec. 3.3 (+8.09 Overall, +21.2 CDM, zero added parameters); the largest model contribution, the layout detector with native reading order (Sec. 3.2), adds +2.22 while halving reading-order edit distance.

4.7. Verification gates, ablated

Verification-gating recurs at three sites (Sec. 3.1); we ablate the two that alter emitted content on the full v1.6 harness (full system 87.29; –CJK hatch 86.91; –empty-page rescue 87.29; –both 86.92; the CJK hatch fires on 170 of 1549 formula chips across 160 pages, the rescue on 15 of 1651 pages). Removing the CJK escape hatch (Sec. 3.3) costs 0.38 Overall, entirely through simplified-Chinese formula CDM ($77.15 \rightarrow 66.76$); English CDM is bit-identical, confirming the gate fires only where Texo is out of distribution. Removing empty-page rescue (Sec. 3.6) leaves the composite unchanged: it is a robustness guard against catastrophic empty output rather than a score component, and fires only where assembly would otherwise return nothing. Disabling both is additive to rounding, and table TEDS is 82.15 in all four arms: neither gate touches structure.

4.8. Emission convention in the composite

A perturbation audit (Supplementary) quantifies how much of the composite reflects emission convention rather than parsing ability: semantically-null transformations move individual metrics by up to 1.5 points and the composite by up to 0.14 points, comparable to many published component-level leaderboard gaps. We recommend such audits as standard practice when comparing document parsers.

5. Conclusion

We localize the gap between PRISM and systems an order of magnitude larger to three strata, all bounded by recognizer capacity (Suppl. Fig. S1): Chinese-embedded formulas (CDM 76 vs. 90 English even after the escape hatch), handwritten formulas (notes CDM 76 vs. 91 on printed exams), and dense agate tables. A larger recognizer would close these, trading against the budget that defines the system. Within that budget PRISM reaches 87.3 on v1.6 and 89.0 on the v1.5 cut, overtaking 7B GPU VLMs and a self-measured MinerU baseline at $4.8\times$ fewer weights. Larger recognizers remain the correct choice where a GPU is available. We measure how far composition carries a system when it is not. Two limitations bound unattended

deployment: failures concentrate in the strata above with no per-page confidence signal to flag them, and behavior on adversarial or corrupted inputs is uncharacterized. Source, configuration and the experiment log accompany this submission as anonymized supplementary material.

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